

VESTA: Evaluating Time-Budgeted Multimodal Briefing for BIST Retail Users

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Abstract. Retail investors on Borsa Istanbul (BIST) read KAP list text and charts under a short clock, which VESTA evaluates by encoding same-day KAP list teasers as a 16-d bag, reading a 40-bar candlestick together with the OHLCV numbers, and spending a declared 1-, 3- or 10-minute budget by changing how many tokens are shown while the mixer stays fixed. A 2σ realized-vol flag on the same day’s OHLCV is recovered at 100% F1 by a closed-form rule, so the headline task is next-day BIST100 direction. Score-space mixers and a learned Arevalo GMU (W_v, W_t, W_z on raw 16-d text and the 24×24 image) are reported on public BIST100 data (2018 to 2026). No score-space mixer beats score-space GMU on seed 0 (minimum McNemar $p = 0.46$); learned GMU does not either ($p = 0.29$). Five-seed accuracy remains compatible with the majority prior (mean fusion 53.0 ± 2.2 versus 52.3). T1/T3/T10 change coverage and templated length, not next-day accuracy (T3 and T10 both 49.7% on seed 0). A long/short overlay loses to buy-and-hold on the 2025 to 2026 bull test window. Human NASA-TLX responses are not claimed. The replication package accompanies the submission.

Keywords: Econometrics · Computational intelligence · Decision support · Multimodal fusion · Borsa Istanbul

1 Introduction

Retail investors sit between the textual density of the Public Disclosure Platform (KAP) [18] and the visual flux of technical charts. The difficulty is not only volume but modality mismatch: a constructive disclosure can coincide with a failed breakout, and the two streams must be reconciled before a decision. Conventional DSS pipelines keep NLP and computer vision apart, and a second failure mode is temporal, because systems emit the same information density whether the user has sixty seconds or ten minutes [7, 17, 22]. On a mobile-first retail platform that mismatch adds intrinsic load [23].

FinMem layers trading-agent memory by information half-life [27]. VESTA scores a briefing whose clock is 1, 3 or 10 minutes (KAP list text and a 40-bar chart), and the fusion rule does not change with the clock: only how many tokens

are shown. T3 and T10 attach an M3 day-level hop, and that analogue does not enter the mixer. Decision support is scored on next-day BIST direction rather than on a realized-volatility flag that can be read off the same day’s OHLCV table. Whenever a chart encoder is used, a numeric OHLCV baseline is reported beside it [11]. The public tables mix unimodal class scores with GMU [2], tensor fusion [28] and a scalar gate, and they add one learned-GMU row that fits Arevalo’s W_v, W_t, W_z on the raw vectors. Actionability is a long/short overlay against buy-and-hold. The public chart is bars $t - 40, \dots, t - 1$. A 12-participant NASA-TLX protocol is specified in `study/`, and responses are not collected.

T1, T3 and T10 change how many tokens are shown while the mixer stays fixed. Because a closed-form rule recovers the same-day 2σ realized-vol flag at 100% F1, the headline label is next-day direction. Tiers report coverage, declared latency, tokens and length-normalized noise and do not raise next-day accuracy. The action metric is a long/short overlay against buy-and-hold on the 2025 to 2026 bull window; Table 8 is a 2023 holdout on the same index and does not reverse that claim. Score-space mixers are not separable from score-space GMU on seed 0 (minimum McNemar $p = 0.46$), and learned GMU is not either ($p = 0.29$). Harvey et al. [10] treat an uncorrected search over many rows as a control rather than a ranking.

1.1 Related work

Financial NLP has moved from sentiment scoring [20] to domain LLMs such as FinGPT [25]. Hallucination on prices remains a documented failure mode [13], so public text here is KAP list teasers rather than free generation [15]. Tool-using agents [26] and FinMem [27] are the closest named relatives, and this paper scores a briefing rather than a trading agent. The public hop is M3 day-level cosine rather than a portfolio retrieve-then-generate loop.

1.2 Multimodal fusion and chart understanding

GMU learns a multiplicative gate between modalities [2]; Tensor Fusion Networks use an outer product [28]; MulT uses directional pairwise cross-modal attention [24]. Jiang and Ji add a forget-gate over cross-modal attention maps [12]. Public tables mix unimodal class scores with GMU, TFN, a scalar gate, mean, concat and one CPU attention layer. That layer is not the MulT of Tsai et al.: projections are frozen and there is a single head. A learned-GMU row fits Arevalo’s maps on the raw vectors.

Pix2Struct and MatCha pretrain screenshot-to-text models [14, 16]. Jiang, Kelly and Xiu show that CNNs on price-trend images extract return-predictive patterns [11]. That result requires a tabular OHLCV baseline beside any chart encoder. Akturk et al. forecast BIST100 from numeric indicators [1]; those features are the control whenever a chart encoder is scored.

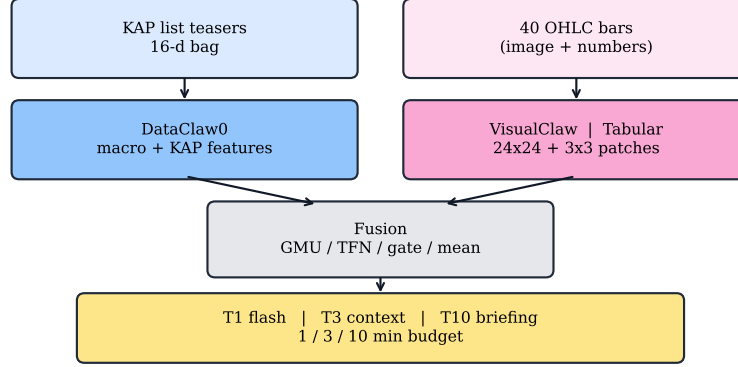


Fig. 1. Public path: KAP list bag and a 40-bar chart go through a mixer; T1, T3 or T10 spends the 1-/3-/10-minute budget.

1.3 Cognitive load

Presented density should match working-memory limits [22, 23, 17, 7]. Adaptive interfaces are judged by the task rather than by whether a widget changed [9, 8]. Length-blind summarization metrics reward brevity [21], which is why Table 6 reports IN_{new} on delivered tokens as a payload count rather than a human-load result. The NASA-TLX protocol is Sect. 2.6, and this manuscript has no responses.

2 Methods

Methods cover the pipeline, the mixers, time-budgeted delivery, and the evaluation design. VESTA has three layers (Fig. 1). Perception uses a text module and a chart module; an off-the-shelf block mixes them; a temporal layer spends the user’s time budget.

2.1 DataClaw0: text brief

The public mixer path is a 16-d macro+KAP bag plus a templated brief. The retrieve-score hop is executed on cached BGE-M3 KAP-day vectors [3, 15, 26] with $\alpha = 1$, unused β gold / USD/TRY / session terms, and no user portfolio.

For each test day the hop is the prior KAP day with the highest cosine to that day’s M3 vector. A same-date XU100 constituent with a KAP teaser is recorded as a peer hop. The retrieved analogue does not enter the 16-d mixer. T3 and T10 still spend a `hist_analog` token; the console attaches the retrieved date and teaser.

The text dataset is VESTA-Public: one row per ticker-day (37,046 events; 27 BIST names plus XU100; 24 May 2018 to 19 August 2026). A row is kept

Table 1. Example KAP list teasers in VESTA-Public (subject plus short summary from kap.org.tr), in calendar order. KAP pol. is kap_polarity, the KAP-only silver codebook, not mixed text_polarity.

Date	Ticker	KAP list teaser	KAP pol.
2018-05-25	YKBNK	Özel Durum Açıklaması (Genel). Bankamıza verilen idari para cezasının iptaline ilişkin açılan dava hk.	bearish
2018-06-14	BIMAS	Payların Geri Alınmasına İlişkin Bildirim. 13 Haziran 2018 Tarihli Pay Geri Alım İşlemleri	bullish
2018-06-19	ASELS	BISTECH Pay Piyasası Alım Satım Sistemi Duyurusu. Temettü Ödemesi	bullish
2020-08-11	FROTO	Özel Durum Açıklaması (Genel). Rekabet Kurulu Soruşturma Kararı Hakkında	bearish

if the name moved at least 2%, the vol flag fired, RSI was extreme, or a KAP list item was published that calendar day. 19,645 rows have a same-day KAP teaser; 15,212 have a cached public HTML body. Yahoo headlines exist for 41 rows. Table 1 lists four KAP strings in calendar order. A Turkish/English lexicon plus subject priors (dividend and buyback tilt bullish; fine and probe tilt bearish) yields kap_polarity. The same strings, plus USD/TRY, gold and the session, become an eight-plus-eight feature vector for the mixers. MiniLM [19] and BGE-M3 embed the concatenated list text. The silver event-row brief for BIMAS on 2018-06-14, rebuilt from the stored columns, is: “2018-06-14 BIMAS.IS. Session return +6.33%. USD/TRY +1.65%, gold +0.55%. Ann. realized vol 33.8%, RSI 48. Brief polarity bullish; chart cue none. Payların Geri Alınmasına İlişkin Bildirim. 13 Haziran 2018 Tarihli Pay Geri Alım İşlemleri”. That string is the ticker-day record, not the T10 token bag.

2.2 VisualClaw: candles, pixels, tokens

VisualClaw draws the chart from OHLC bars. Figure 2 is one XU100 window from the test split.

1. Take the 40 sessions strictly before day t (open, high, low, close). Today’s candle is not in the picture.
2. Paint a grayscale candlestick: a one-pixel wick and a rectangular body. Close $\geq$ open is dark (30/255); close $<$ open is light (180/255). Price is min-max scaled to that window.
3. Resize to 24×24 . The vision MLP flattens this to a 576-d vector and classifies with hidden layers 64-32. That is the vision row in Table 4.
4. For the 1-layer attention mixer only, cut the same 24×24 image into a 3×3 grid of 8×8 patches (nine visual tokens). The 16-d text brief is split into two tokens. Projections are frozen (seed 0); only the MLP on top is trained.

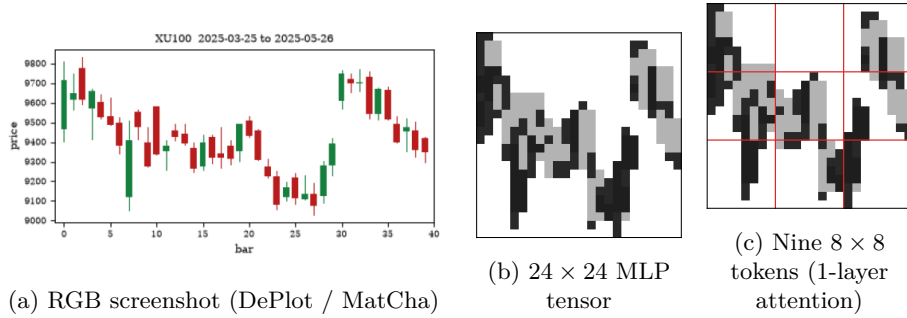


Fig. 2. VisualClaw on one XU100 test window (40 bars before 27 May 2025). (a) RGB screenshot for DePlot / MatCha. (b) 24×24 grayscale tensor the MLP flattens. (c) Nine 8×8 tokens for the 1-layer attention mixer.

Two further probes are reported with the mixer: a frozen ImageNet ViT-B/16 [6] that resizes a colour chart to 224×224 and uses that model’s own 16×16 patches, and zero-shot DePlot / MatCha, which see a matplotlib RGB screenshot of the same 40 bars (Fig. 2(a)). Tabular OHLCV at day t is always reported beside the picture [11]. Public tables train the flattened 24×24 MLP; they do not add a recency term to patch positions.

2.3 Fusion

Arevalo et al. write the bimodal GMU as [2]

$$h_v = \tanh(W_v x_v), \quad h_t = \tanh(W_t x_t), \quad (1)$$

$$z = \sigma(W_z [x_v; x_t]), \quad h = z h_v + (1 - z) h_t, \quad (2)$$

so z near 1 keeps the visual branch. Zadeh et al. flatten $[h_t; 1] \otimes [h_v; 1]$ (TFN) [28]. Table 4 has two GMU rows. The score-space GMU takes unimodal class-score vectors as x_t and x_v and sets z from the score gap (Arevalo’s mix; W_v, W_t, W_z unused in that row only). The learned-GMU row fits those three maps on the raw 16-d text vector and the flattened 24×24 image by minibatch gradient descent, then a linear head on h . The TFN row uses the outer product on the same class scores. The scalar-gate row uses

$$g = \sigma(4(p_{\text{text}} - p_{\text{vision}})), \quad (3)$$

$$h = g \tanh(p_{\text{text}}) + (1 - g) \tanh(p_{\text{vision}}), \quad (4)$$

so g near 1 keeps text. A small MLP reads the mix. Mean and concat are unweighted. The 1-layer attention row is one CPU attention layer on two text tokens and nine 8×8 patches, with frozen projections. That layer is not Tsai MulT [24]. The fused call drives T1 / T3 / T10.

2.4 Temporal Orchestration Layer

The design factor is the user’s clock $B \in \{1, 3, 10\}$ minutes; no participant was timed against that cap, and Table 6 reports declared latencies 0.9 / 2.1 / 3.4 s rather than wall-clock. T3 and T10 emit the same fused call, and extra tokens are templates.

- T1 (flash) emits a gated alert only if $|g-0.5| > \tau$ with $\tau = 0.18$ and otherwise abstains, with declared latency 0.9 s.
- T3 (context) always emits the fused call plus a retrieved analogue token (M3 cosine, $\alpha = 1$, no portfolio).
- T10 (briefing) expands into a macro explanation and a hedge sketch (e.g. airline exposure versus a gold certificate). The 3.4 s figure in Table 6 is a declared budget rather than measured wall-clock.

The mixer stays fixed while tiers change how many tokens are shown. Section 3.5 reports coverage, latency, tokens and noise per tier.

2.5 Evaluation protocol

The evaluation has a leakage diagnostic, a primary next-day direction task, per-tier delivery metrics, and a paper-trading overlay.

Diagnostic: the realized-volatility label The 2σ realized-vol flag at day t is 20-day realized vol (annualized) above the 60-day rolling mean plus 2σ of that 20-day vol series, all taken from the tabular vector at t (100% F1). The public chart does not contain today’s candle, so vision F1 on this label is lag-1 recovery. We report the flag only as a diagnostic.

Primary task: forward direction The primary label at close of day t is the next session’s direction: $y_{t+1} = 1$ if the BIST100 simple return r_{t+1} is positive, and $y_{t+1} = 0$ otherwise. The label is unknown at the close of day t . We also report a five-session horizon. Splits are chronological (70/15/15) to block look-ahead [1].

Metrics Classification uses accuracy and macro-F1, mean ± 1.96 standard error over five seeds, and McNemar with continuity correction [5] on seed 0 of one chronological holdout. Table 4 has fourteen rows. Pairwise McNemar versus score-space GMU is a control (minimum $p = 0.46$ for score-space mixers; $p = 0.29$ for learned GMU). With that many rows we follow Harvey et al. [10] and do not pick a winner after an uncorrected search.

Table 2 in Sect. 3 reports proxy accuracy versus next-day direction. Macro flags and KAP polarity are deterministic rules on one chronological split (no seed CI). Vision and tabular are five-seed means from Table 4. Three KAP codebooks on the 10k slice give Fleiss $\kappa = 0.50$; there are no human annotators. Chart A/B

Cohen $\kappa = 0.52$ [4]. The 250-row sample has no annotators, and those columns are codebook B.

Information noise is the computational cognitive measure. Let N_{raw} be tokens in the unprocessed window (the raw KAP list plus session flags) and N_{del} tokens actually delivered in the T1, T3 or T10 briefing. Let N_{irrel} be delivered tokens that are not in a relevance lexicon for that window (session, USD/TRY, gold, vol, KAP digest). The index $\text{IN}_{\text{old}} = N_{\text{irrel}}/N_{\text{raw}}$ rewards short answers: a 40-token briefing that is 80% filler still looks clean against a 700-token feed [21]. We therefore report

$$\text{IN}_{\text{new}} = N_{\text{irrel}}/N_{\text{del}}, \quad (5)$$

the fraction of the delivered finance briefing that is non-actionable, together with the compression ratio $N_{\text{del}}/N_{\text{raw}}$. IN_{new} is our index; Sun et al. motivate dividing by delivered length. Both counts use templated token bags with filler padding, so IN_{new} rises by construction when T3/T10 add filler. Table 6 is that computational proxy; human NASA-TLX is unreported (Sect. 2.6).

Classification F1 does not say whether acting on the output helps [8]. We paper-trade a long/short overlay: +1 if the model predicts up, −1 otherwise, with 5 bp round-trip cost, and compare Sharpe, total return and max drawdown to buy-and-hold. Table 7 is that control; Sect. 2.6 is the human cognitive protocol.

Public replication corpus Numbers below use VESTA-Public: XU100.IS, USDTRY and COMEX gold from 24 May 2018 to 19 August 2026 (2,052 index windows after a 40-day lookback; test $n = 308$ from 27 May 2025), 27 BIST names, and the public KAP list inventory of 39,890 filings (27 names; scrape window 2 January 2018 to 27 August 2026). Tables 2 to 7 are index-level; the 37,046-row event panel also labels constituents. KAP-only polarity: 7,172 / 3,742 / 26,132 (bullish / bearish / neutral); 15,212 rows have a cached public HTML body. Mixers use the 16-d macro+KAP vector; MiniLM and BGE-M3 are unimodal text probes on the same list teasers.

2.6 Human protocol

Computational load is reported with the tier table in Sect. 3.5. A BIST retail user on a mobile desk has about one, three or ten minutes between a KAP drop and an order, and that clock is the independent variable. The dependent quantities are coverage (how often the system speaks), declared latency, tokens delivered, and IN_{new} (what fraction of the delivered briefing is non-actionable). The token bags are built from the day’s finance flags: session return, USD/TRY, gold, realized vol, a KAP digest, and, on T3/T10, a historical analogue and a hedge sketch. Those counts measure load on the payload rather than load on a person [9, 8].

The human protocol is in **study/**. Twelve participants would see eight sealed snapshots from the public test window (two XU100 sessions and six constituents: SASA, SISE, YKBNK, TAVHL, AKBNK, PETKM), within subject. The sealed gold is the subsequent week’s realized direction of that name. Conditions are the

Table 2. Proxy accuracy versus next-day direction y_{t+1} (test $n = 308$). Macro/KAP are single-split rules; vision/tabular are mean ± 1.96 se over five seeds.

Proxy	Acc. (%)
Macro flags (USD/TRY, gold, vol, shock)	54.9
Vision 24×24 MLP	52.9 ± 2.3
Tabular OHLCV	50.6 ± 0.0
KAP polarity (lexicon on public list text)	49.7

raw KAP+OHLC feed, a unimodal text or chart, or the matching VESTA tier, under a 1-, 3- or 10-minute cap. Outcomes are the week-direction call, time-to-decision, NASA-TLX, and a 7-point trust item. Consent and the bilingual instrument are in `study/`.

Responses have not been collected ($n = 0$), so human-load and trust scores are unreported. A model dry-run on the sealed gold hits 4/8 name-level weeks for the raw session sign; index mixers are scored only on the two XU100 weeks (text 0/2, gated 1/2). NASA-TLX and trust stay blank. The executable protocol remains in `study/` so a later sample can be scored against the same gold.

3 Results

Unimodal encoders are sklearn MLPs (64-32, Adam, early stopping) on standardized features. Vision sees a 24×24 grayscale candlestick of bars $t - 40, \dots, t - 1$; tabular sees eleven OHLCV/macro features at t , including the ingredients of the leaked vol rule; text sees the sixteen-dimensional macro+KAP vector. Five seeds $\{0, \dots, 4\}$. Training is on CPU. Public BGE-M3 and frozen ViT-B/16 are unimodal probes; they remain unimodal in Table 4.

3.1 Proxy accuracy

Table 2 is the rule and encoder proxy against next-day direction. Macro flags are the strongest single-split rule (54.9%). KAP polarity on list teasers is 49.7%. The 24×24 vision MLP and tabular OHLCV repeat their five-seed means from Table 4.

3.2 Diagnostic: the leaked volatility label

Table 3 and Fig. 3 confirm the closed-form critique. The 2σ rule on the tabular vector scores 100% F1. A tabular MLP that is given those numbers reaches 96.8% accuracy (macro-F1 86.6%; the F1 drop is class imbalance: 25 positives among 308 test days, 8.1%; 9.4% of all 2,052 windows). The 24×24 vision encoder, which never sees day t 's candle, falls to 90.6% accuracy and 58.2% F1.

Table 3. Diagnostic task: 2σ realized-vol label, computable from the tabular vector at t . The closed-form row is a single split. MLP rows are mean ± 1.96 se over five seeds. Vision sees bars $t - 40, \dots, t - 1$ only.

Method	Accuracy (%)	Macro-F1 (%)
Closed-form OHLCV rule	100.0	100.0
Tabular MLP (same numbers)	96.8 ± 0.0	86.6 ± 0.0
Vision MLP (chart image)	90.6 ± 1.1	58.2 ± 3.0

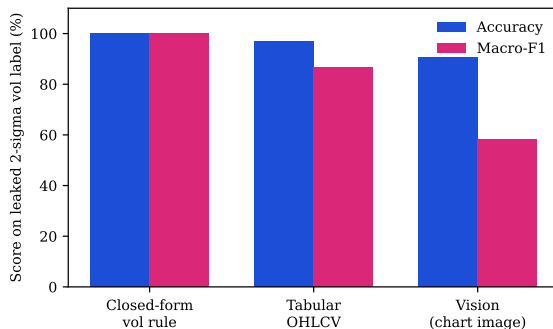


Fig. 3. Diagnostic 2σ -vol label. The closed-form OHLCV rule scores 100% F1; the chart omits day t .

3.3 Primary task: next-day direction

Five-seed accuracy CIs overlap the majority prior 52.3%, and mean fusion 53.0 ± 2.2 does not separate from that prior. Table 4 reports the leakage-free task (test prior 52.3% up-days). Accuracy / macro-F1: text-only (macro+KAP) 52.1 ± 2.4 / 47.3 ± 1.5 against majority 52.3 / 34.3; MiniLM 53.0 ± 0.4 / 50.7 ± 1.9 ; public BGE-M3 51.1 ± 1.9 / 50.2 ± 2.1 ; frozen ViT-B/16 50.9 ± 1.4 / 50.3 ± 1.6 . Mixers share 16-d text and the 24×24 image. Among score-space mixers, none beats score-space GMU on seed 0 (minimum McNemar $p = 0.46$). Learned GMU (W_v, W_t, W_z on the raw vectors) is 52.5 ± 0.5 / 40.3 ± 4.4 and does not beat score-space GMU either (seed-0 McNemar $p = 0.29$); the F1 drop is majority collapse. On the five-session horizon (seed 0) tabular accuracy / F1 is 59.1% / 39.9%; vision is 51.9% / 49.0% and the scalar gate is 50.3% / 49.3%.

Replacing the gate with a fixed mean changes five-seed F1 from 51.4 ± 1.8 to 52.4 ± 2.3 (seed-0 McNemar $p = 1.00$).

3.4 Zero-shot Pix2Struct and MatCha

Zero-shot DePlot [14] and MatCha-ChartQA [16] run on CPU matplotlib screenshots of the full test window ($n = 308$), with no fine-tune. Calls are $\text{sign}(\hat{y}_{\text{last}} - \hat{y}_{\text{first}})$. Table 5 also scores the numeric window trend of the input OHLC, which

Table 4. Next-day BIST100 direction on VESTA-Public (test $n = 308$). Mean ± 1.96 se, five seeds. MiniLM, BGE-M3 and ViT are unimodal probes. Mixers include 1-layer attention, GMU [2] and TFN [28]. Score-space GMU/TFN/gate mix unimodal class scores. Learned GMU fits Arevalo’s W_v, W_t, W_z on raw text and the 24×24 image. The 1-layer attention row is a single frozen-projection head, not Tsai et al.’s MulT.

Configuration	Accuracy (%)	Macro-F1 (%)
Mean fusion	53.0 ± 2.2	52.4 ± 2.3
Text-only (MiniLM KAP)	53.0 ± 0.4	50.7 ± 1.9
Vision-only (24×24 MLP)	52.9 ± 2.3	51.4 ± 2.6
Concatenation	52.8 ± 1.8	51.1 ± 2.3
Learned GMU (W_v, W_t, W_z)	52.5 ± 0.5	40.3 ± 4.4
Majority class	52.3 ± 0.0	34.3 ± 0.0
TFN	52.1 ± 2.1	51.7 ± 2.1
Text-only (macro + KAP)	52.1 ± 2.4	47.3 ± 1.5
Scalar gate (score-space)	51.8 ± 1.7	51.4 ± 1.8
Text-only (public BGE-M3)	51.1 ± 1.9	50.2 ± 2.1
Vision-only (frozen ViT-B/16)	50.9 ± 1.4	50.3 ± 1.6
GMU (score-space)	50.6 ± 2.4	50.6 ± 2.4
Tabular OHLCV	50.6 ± 0.0	41.8 ± 0.0
1-layer attention	50.2 ± 3.1	48.4 ± 3.0

Table 5. Zero-shot CPU screenshot VLMs on the full test window ($n = 308$), with no fine-tune.

Method	Acc. vs y_{t+1}	Macro-F1	Acc. vs window trend
DePlot (Pix2Struct)	52.9	51.9	77.9
MatCha-ChartQA	52.3	46.0	58.4
Majority	52.3	34.3	n/a
Numeric window trend	51.6	49.5	100.0

is still not y_{t+1} . DePlot parses 307/308 images and recovers that trend on 77.9% of days; next-day accuracy is 52.9%. MatCha-ChartQA often returns the bar index (parse rate 166/308) and matches the majority class on y_{t+1} .

3.5 Per-tier measurement

Table 6 and Fig. 5 measure the Temporal Orchestration Layer. Latencies 0.9 / 2.1 / 3.4s are declared budgets rather than wall-clock, and no participant was timed against the 1-/3-/10-minute cap. On seed 0, T1 covers 32.5% of days (60.0% among emitted calls; 55.2% after majority fill). T3 and T10 always emit (coverage 100%). T3 accuracy equals T10 accuracy (both 49.7%) because extra tokens are templates, not new evidence. The M3 hop retrieves on all 308 test days (mean cosine 0.89). The retrieved day’s next-day sign matches y_{t+1} on 49.8% of 305 scored days. That analogue does not beat the majority prior, and it does

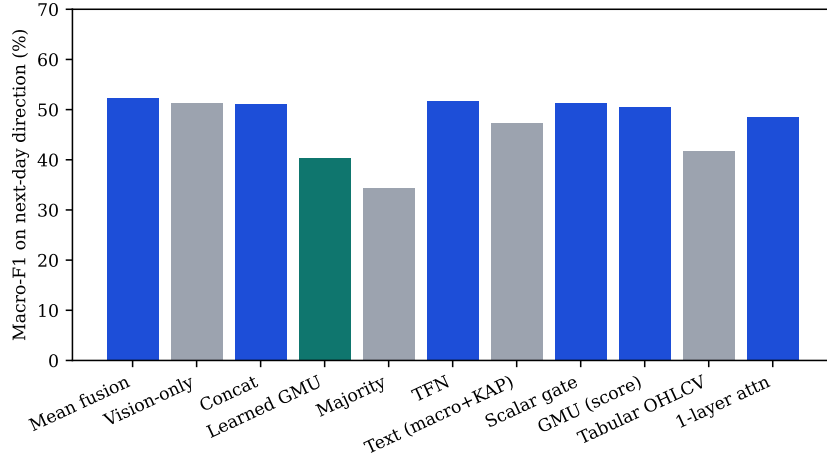


Fig. 4. Macro-F1 for score-space mixers, sklearn unimodal rows, and learned GMU from Table 4, in the same row order. MiniLM, BGE-M3 and ViT stay in the table only. Mean ± 1.96 se is in the table.

Table 6. Temporal Orchestration Layer on seed 0. Latency is a declared budget, not measured GPU time. Acc. is among emitted calls; T1 overall accuracy with majority fill is 55.2%.

Tier	Cov. (%)	Lat. (s)	Acc. (%)	Tokens	IN _{old}	IN _{new}
T1 flash	32.5	0.9	60.0	6	0.5%	60.3%
T3 context	100.0	2.1	49.7	17	1.9%	82.1%
T10 briefing	100.0	3.4	49.7	52	6.8%	93.6%

not enter the mixer. $IN_{old} < 7\%$ for every tier. IN_{new} is 60.3 / 82.1 / 93.6%; that series is the padding index. It rises by construction when T3/T10 add filler.

3.6 Decision quality

Table 7 reports the paper-trading overlay. In the May 2025 to August 2026 test window, BIST100 buy-and-hold returned +56.3% (Sharpe 1.69). Every seed-0 overlay we tried underperformed that long-only benchmark after 5 bp costs. Vision-only (Sharpe 0.37) and the scalar gate (0.33) were positive but far below BH; GMU and tabular were negative. A bull-market holdout is a hard test for a long/short overlay [10].

3.7 Earlier holdout (2023)

Table 8 is a second chronological cut on XU100. Models are trained on days strictly before 2 January 2023 and scored on 2023 ($n = 248$; prior 51.2%).

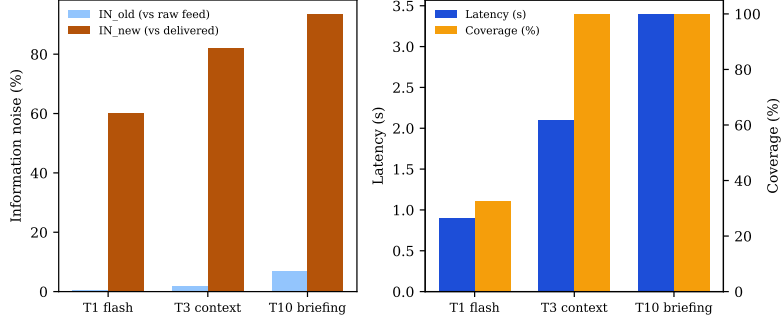


Fig. 5. Left: length-normalized noise of the delivered briefing (IN_{old} versus IN_{new}). Right: declared latency and coverage for T1, T3 and T10.

Table 7. Paper-trading overlay on seed-0 calls, one bull chronological window (27 May 2025 to 19 August 2026), 5 bp costs. BH is long-only BIST100. Dir. is the fraction of days the signed position matches y_{t+1} ; for buy-and-hold that equals the up-day prior. Long/short versus BH is a hard test on this window. Table 8 is the earlier 2023 holdout on the same index.

Policy	Dir. (%)	Sharpe	Total ret.	Max DD
Buy-and-hold	52.3	1.69	+56.3%	14.5%
Vision-only overlay	50.0	0.37	+7.6%	16.4%
Scalar-gate overlay	49.7	0.33	+6.3%	17.1%
Tabular overlay	50.6	-0.11	-6.2%	23.0%
GMU overlay	48.1	-0.74	-21.8%	27.4%

Five-seed mean fusion accuracy is 53.3 ± 1.7 against majority 51.2. On seed 0 the vision overlay beats buy-and-hold (Sharpe 1.81 versus 0.99). That does not reverse Table 7: two uncorrected windows are not a ranking [10].

4 Discussions

The public path runs an M3 day-level hop ($\alpha = 1$, no portfolio). The analogue hit rate is 49.8%, and the hop does not enter the mixer. T3 and T10 both score 49.7% on seed 0 because extra tokens are templates. Five-seed accuracy remains compatible with the majority prior. The 2025 to 2026 bull overlay loses to buy-and-hold, and the 2023 holdout does not reverse that claim. Human NASA-TLX is unreported ($n = 0$). Fleiss $\kappa = 0.50$ is three codebooks; there are no human annotators. Public text is KAP list teasers rather than full KAP HTML.

Learned GMU does not beat score-space GMU ($p = 0.29$), and no score-space mixer does either (minimum $p = 0.46$). The learned-GMU row is the control for fusing unimodal class scores. The public chart uses bars $t - 40, \dots, t - 1$. Information noise and tier latencies are simulated. IN_{new} rises by construction when T3/T10 add filler.

Table 8. Earlier holdout on XU100, 2 January 2023 to 29 December 2023 ($n = 248$). Train is all sessions before 2023-01-01 ($n = 1149$). Accuracy is mean $\pm$ 1.96 se over five seeds. Overlay Sharpe and return are seed 0, 5 bp costs; n/a where that mixer has no overlay row. Buy-and-hold on the same window: Sharpe 0.99, +34.7%.

Configuration	Acc. (%)	Macro-F1 (%)	Sharpe	Total ret.
Mean fusion	53.3 ± 1.7	51.3 ± 2.8	n/a	n/a
Vision-only (24×24 MLP)	53.1 ± 1.8	50.6 ± 2.9	1.81	+81.4%
Scalar gate (score-space)	52.5 ± 1.7	50.4 ± 1.2	1.64	+70.6%
Tabular OHLCV	51.2 ± 0.0	34.6 ± 0.0	0.56	+14.8%
Majority class	51.2 ± 0.0	33.9 ± 0.0	n/a	n/a
Text-only (macro + KAP)	50.4 ± 1.3	40.5 ± 2.3	n/a	n/a

The 12-participant NASA-TLX study in `study/` has no responses, so human load and trust are unreported. There is no human-annotator gold set; polarity is silver codebook output. Full disclosure-detail pages are absent.

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